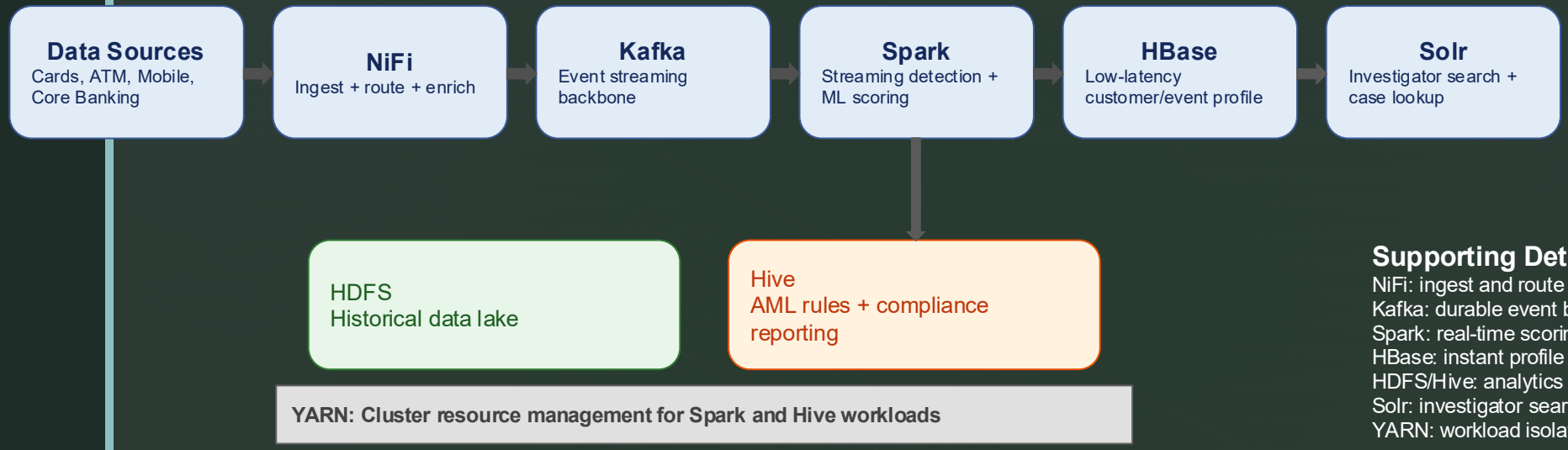
Problem Statement

- Global bank fraud and AML risk is rising across channels and geographies.
  - Unauthorized card transactions and account takeover attempts are increasing in volume and sophistication.
  - Batch-only detection causes delayed response, customer friction, and preventable financial loss.
  - Current monitoring generates high false positives, overwhelming investigators and slowing case resolution.
  - Regulators require explainable alerts, full audit trails, and strong anti-money laundering controls.
  - Business goal: detect and block high-risk events in real time while preserving customer trust and compliance.



Proposed Architecture



Supporting Details

NiFi: ingest and route streams
Kafka: durable event buffering
Spark: real-time scoring engine
HBase: instant profile lookups
HDFS/Hive: analytics + AML evidence
Solr: investigator search & triage
YARN: workload isolation + scaling



Benefits & Conclusion

- **Business and Technical Benefits**
- Real-time fraud response: transactions are scored and risky events are blocked within seconds.
- Higher detection quality: behavior analytics reduce false positives and improve investigator focus.
- Regulatory readiness: Hive-based AML reporting and lineage support auditability and transparency.
- Scalable platform: Kafka + Spark + YARN support growth in volume without redesign.
- Resilient operations: distributed storage/compute increase availability and reduce single points of failure.
- Customer trust and brand protection improve as fraud losses and response times decline.
- **Conclusion: This architecture gives the global bank a practical, explainable, and scalable path to fraud prevention and AML compliance.**

